

smaller, which translates to savings in terms of real estate. Operating temperatures are much lower because it is not necessary to heat the process liquid above its boiling point. These benefits result in less heat loss to the environment through the equipment surface. Therefore low-grade waste heat energy source like solar, geothermal energy and waste heat from industrial plants can be coupled with MD system.

In fact, MD plants powered by waste heat of PV cells are cost-competitive with RO process for clean water production. Easier operating conditions, lower safe operating pressure at ambient conditions and lesser fouling problems are other benefits. MD process is a very efficient salt removal system from raw sea water by reducing its input salinity from ~ 25,500mg/l TDS content to 9mg/l TDS.

In case MD-produced water is to be used as drinking water, direct addition of certain minerals that are needed by our body must be added to the clean water as per WHO standards.

The solar cell harvests short wavelength sunlight to generate electricity. Large amount of waste heat is simultaneously generated as a side-effect from two pathways. The first one is relaxation of short wavelength sunlight-excited electrons, and the second pathway is photothermal conversion of long wavelength sunlight. In a typical solar cell, eighty to ninety per cent of absorbed solar energy is undesirably converted to heat. Waste heat is dumped into ambient air as waste. Using this concept, waste heat is considered as a resource and utilised to power PV-MD devices to produce clean water.

Fig. 9 shows the arrangement of an integrated PV-MD device. Commercial polycrystalline silicon solar cell is used both as an electricity generator and as a photothermal component. PV-MD device is constructed at the back of the solar cell for clean water production. To reduce heat loss into the atmosphere, sides of the composite device are sealed by polyurethane foam with low thermal conductivity.

Each stage of the MSMD device has four separate layers:

- Top thermal conduction layer

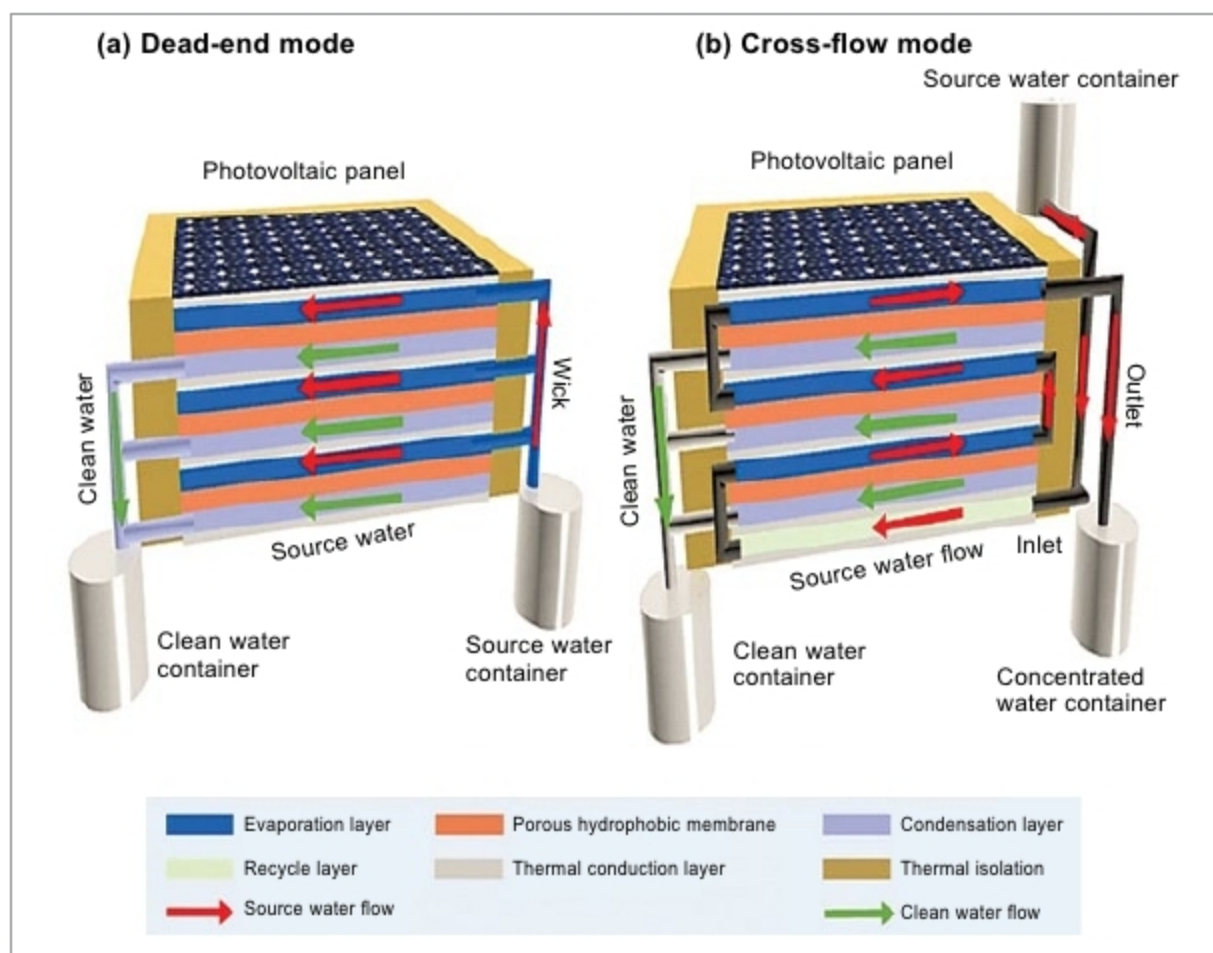


Fig. 9: Arrangement of composite solar cell with membrane distillation

- Hydrophilic porous water evaporation layer
- Hydrophobic porous layer of MD membrane for vapour permeation
- Water vapour condensation layer

Aluminium-nitride (AlN) plate is used as the thermal conduction layer due to its very high thermal conductivity and its anti-corrosive nature to saline water. The hydrophobic porous layer is made from polystyrene (PS) membrane. Water evaporation and condensation layers are composed of the same material—commercial hydrophilic quartz glass fibrous (QGF) membrane with non-woven fabric structure.

In each stage of the MSMD device, heat is conducted through the thermal conduction layer to the underlying hydrophilic porous layer. Source water inside the hydrophilic porous layer is thus heated to produce water vapour. Water passes through the hydrophobic porous membrane layer and ultimately condenses on the condensation layer to produce clean water. The driving force behind water evaporation and vapour condensation is the vapour pressure difference caused by the temperature gradient between evaporation and condensation layers.

In each stage, latent heat of water vapour, which is released during condensation process, is utilised as the

heat source to drive evaporation in the next stage. The multi-stage design ensures that heat can be repeatedly re-used to drive the water evaporation and condensation cycle, which sets up an upper theoretical ceiling for the clean water production rate of ~ 1.60kg/m² per hour, under one Sun condition in such a system.

The system can be operated either in dead-end mode, as shown in Fig. 9(a), where all fluids flow in parallel. Here, source water is passively wicked into the evaporation layer by hydrophilic QGF membrane strips via capillary effect. In this case, salt concentration in the evaporation layer keeps increasing till it reaches saturation. But this mode requires washing to remove accumulated salts inside the device.

The system can also be operated in cross-flow mode, as shown Fig. 9(b), where source water flows into the device driven by gravity or by a pump, and flows out of the device before reaching saturation. Here, outgoing water flow takes away a small amount of sensible heat, leading to a slight drop in clean water productivity in the earlier stage. However, it solves the problem of salt accumulation and avoids the need for frequent cleaning, which makes the device suitable for continuous long-term operation. **EFY**